Growth of Complexity: Search and Sort

Motivating question: Why do some algorithms still feel effortless as data grows, while others suddenly become unusable?

Learning goals - connect the idea of input size `n` to concrete list-processing tasks - recognize the visual shape of common growth laws - compare theoretical operation counts with measured runtime - use search and sorting to see why preprocessing can change the whole story

This notebook accompanies Lecture 03 on computational complexity. It is designed for live demo and self-study.

```
from pathlib import Path
import sys
import math

# The notebook may be launched from the lecture folder or from the repo root.
# We walk upward until we find the shared helper module.
ROOT = Path.cwd().resolve()
for candidate in [ROOT, *ROOT.parents]:
   if (candidate / 'utilities' / 'complexity_utils.py').exists():
      ROOT = candidate
      break
else:
   raise RuntimeError('Could␣not␣locate␣utilities/complexity_utils.py')

if str(ROOT) not in sys.path:
   sys.path.append(str(ROOT))

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from IPython.display import Markdown

from utilities.complexity_utils import (
   display_wrapped,
   COLORS,
   instructor_prompt,
   lecture_figure,
   markdown_box,
   plot_binary_search_demo,
   plot_insertion_sort_passes,
   plot_linear_search_demo,
   plot_merge_sort_decomposition,
   plot_preprocessing_workflow,
   setup_lecture_style,
   time_size_series,
)

setup_lecture_style()
rng = np.random.default_rng(7)
pd.options.display.float_format = lambda value: f'{value:,.4f}'
```

This notebook follows the shared Lecture 03 visual rhythm: a motivating question, a small model, one dominant visual, a compact comparison, and a short takeaway.

This notebook opens the Lecture 03 sequence. It establishes the visual language for the series: start from a motivating question, use one dominant visual per section, and connect asymptotic growth to small measured experiments without confusing the two.

1. What does growth mean?

In this notebook, `n` will usually mean the **number of elements in a list**. That is a natural input measure for search and sorting.

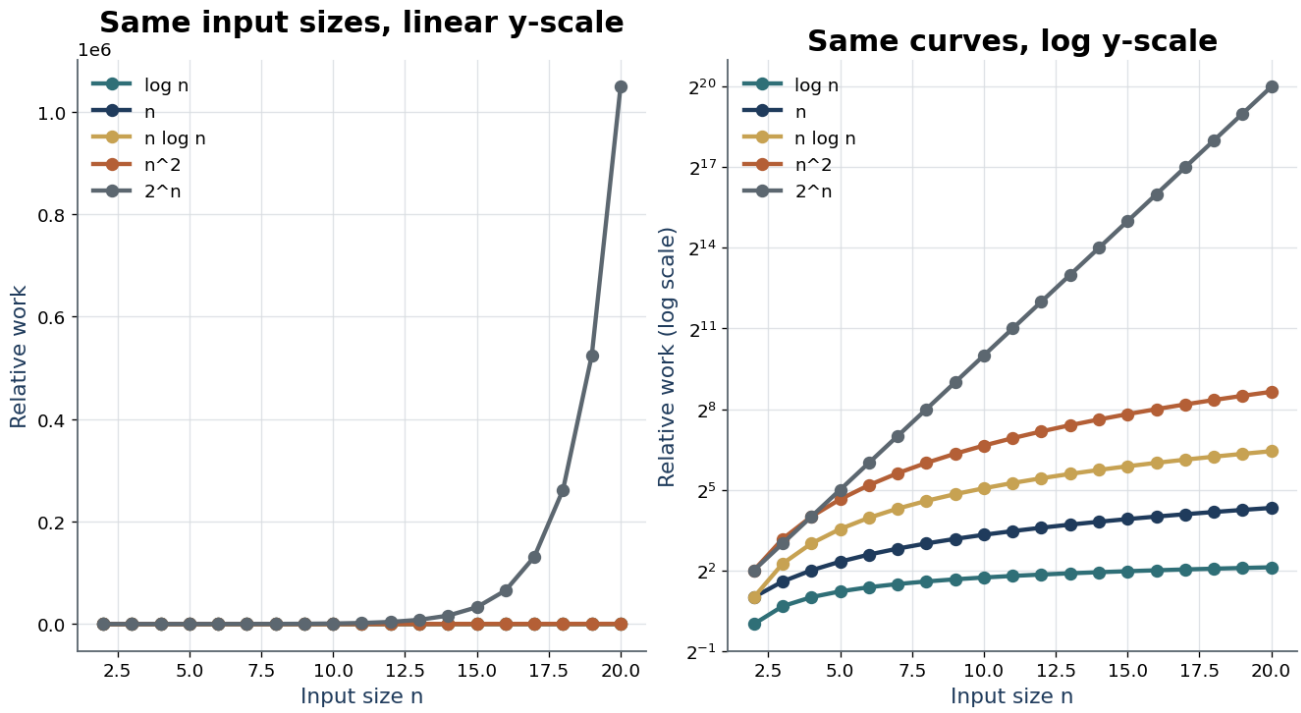
The larger lesson is broader: complexity always depends on how we measure the input. For bit strings, graphs, matrices, or quantum states, the right notion of `n` may be different.

We begin with the simplest visual question: if n grows, which formulas stay manageable and which separate rapidly? The model is intentionally small because the lecture goal is to train the eye before moving to concrete algorithms.

```
# Here n means "list length". We keep n fairly small in the table so the
```

```
# values stay readable during live lecture, even though the point is about
# what happens as n continues to grow.
n = np.arange(2, 21)
growth_df = pd.DataFrame(
   {
      'n': n,
      'log2(n)': np.log2(n),
      'n␣log2(n)': n * np.log2(n),
      'n^2': n**2,
      '2^n': 2**n,
   }
)

display_wrapped(growth_df.iloc[[0, 2, 4, 8, 12, 16, 18]].reset_index(drop=True))

fig, axes = lecture_figure(kind='wide', ncols=2)

# The left panel gives the immediate visual shock: exponential growth
# quickly dominates the same input-size range.
axes[0].plot(n, np.log2(n), marker='o', label='log␣n', color=COLORS['teal'])
axes[0].plot(n, n, marker='o', label='n', color=COLORS['navy'])
axes[0].plot(n, n * np.log2(n), marker='o', label='n␣log␣n', color=COLORS['gold'])
axes[0].plot(n, n**2, marker='o', label='n^2', color=COLORS['rust'])
axes[0].plot(n, 2**n, marker='o', label='2^n', color=COLORS['gray'])
axes[0].set_title('Same␣input␣sizes,␣linear␣y-scale')
axes[0].set_xlabel('Input␣size␣n')
axes[0].set_ylabel('Relative␣work')
axes[0].legend(loc='upper␣left')

# The right panel is not a different experiment. It is the same data on a
# log y-axis, which makes slower-growing curves easier to compare.
axes[1].plot(n, np.log2(n), marker='o', label='log␣n', color=COLORS['teal'])
axes[1].plot(n, n, marker='o', label='n', color=COLORS['navy'])
axes[1].plot(n, n * np.log2(n), marker='o', label='n␣log␣n', color=COLORS['gold'])
axes[1].plot(n, n**2, marker='o', label='n^2', color=COLORS['rust'])
axes[1].plot(n, 2**n, marker='o', label='2^n', color=COLORS['gray'])
axes[1].set_yscale('log', base=2)
axes[1].set_title('Same␣curves,␣log␣y-scale')
axes[1].set_xlabel('Input␣size␣n')
axes[1].set_ylabel('Relative␣work␣(log␣scale)')
axes[1].legend(loc='upper␣left')

plt.show()
```

	n	**log2(n)**	**n log2(n)**	**n^2**	**2^n**
0	2	1.0000	2.0000	4	4
1	4	2.0000	8.0000	16	16
2	6	2.5850	15.5098	36	64
3	10	3.3219	33.2193	100	1024
4	14	3.8074	53.3030	196	16384
5	18	4.1699	75.0587	324	262144
6	20	4.3219	86.4386	400	1048576



On the left, exponential growth quickly overwhelms everything else. On the right, the log-scaled axis lets us see that `n`, `n log n`, and `n^2` are also genuinely different, even if they all looked small for tiny inputs.

```
display_wrapped(Markdown(markdown_box(
    'Tiny differences in growth law become dramatic once n gets large. Choosing the right algorithm is often more
        important than shaving a constant factor.'
)))

display_wrapped(Markdown(instructor_prompt(
    'Which curve would you tolerate for n = 10, for n = 1,000, and for n = 1,000,000? '
)))
```

> **Takeaway:** Tiny differences in growth law become dramatic once n gets large. Choosing the right algorithm is often more important than shaving a constant factor.
>
> **Instructor Prompt:** Which curve would you tolerate for n = 10, for n = 1,000, and for n = 1,000,000?

2. Search as first intuition

Searching is a good first example because the task is simple: find a target value in a list.

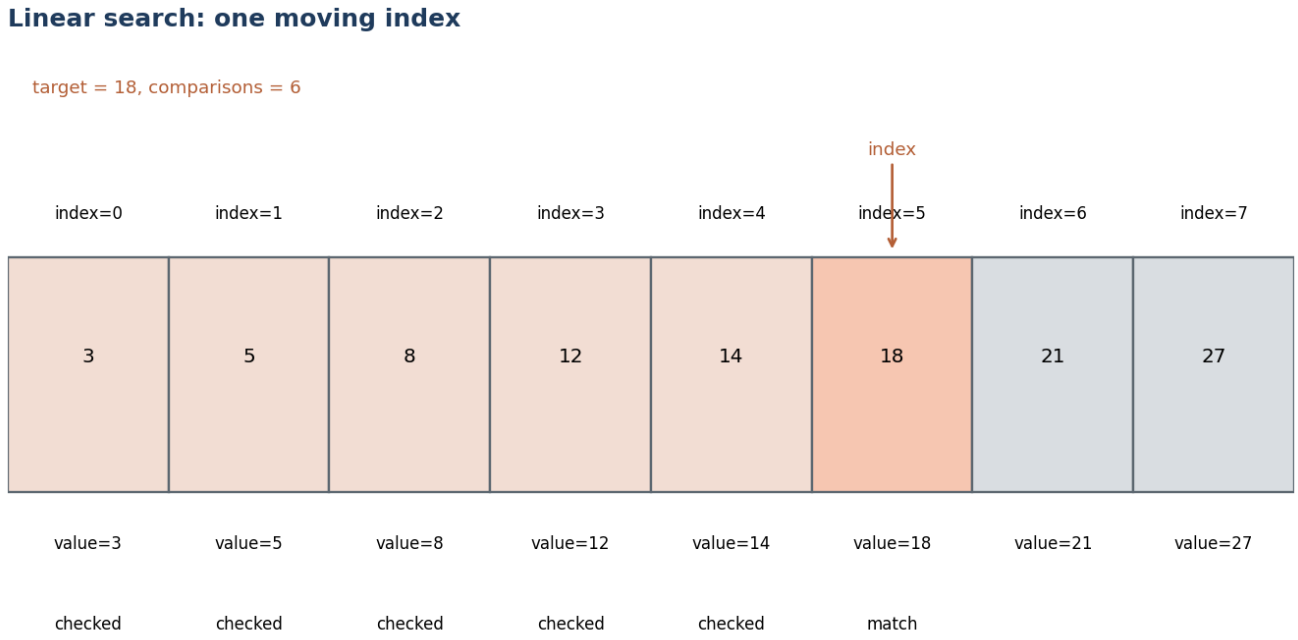
We will compare two ideas: - **linear search** scans one element after another - **binary search** repeatedly cuts the remaining search interval in half

Binary search is much faster in principle, but it only works on **sorted** data.

Search is the first concrete place where scaling intuition becomes operational. We will compare operation counts first, then small measured timings, so students can separate mathematical growth from machine-dependent constants.

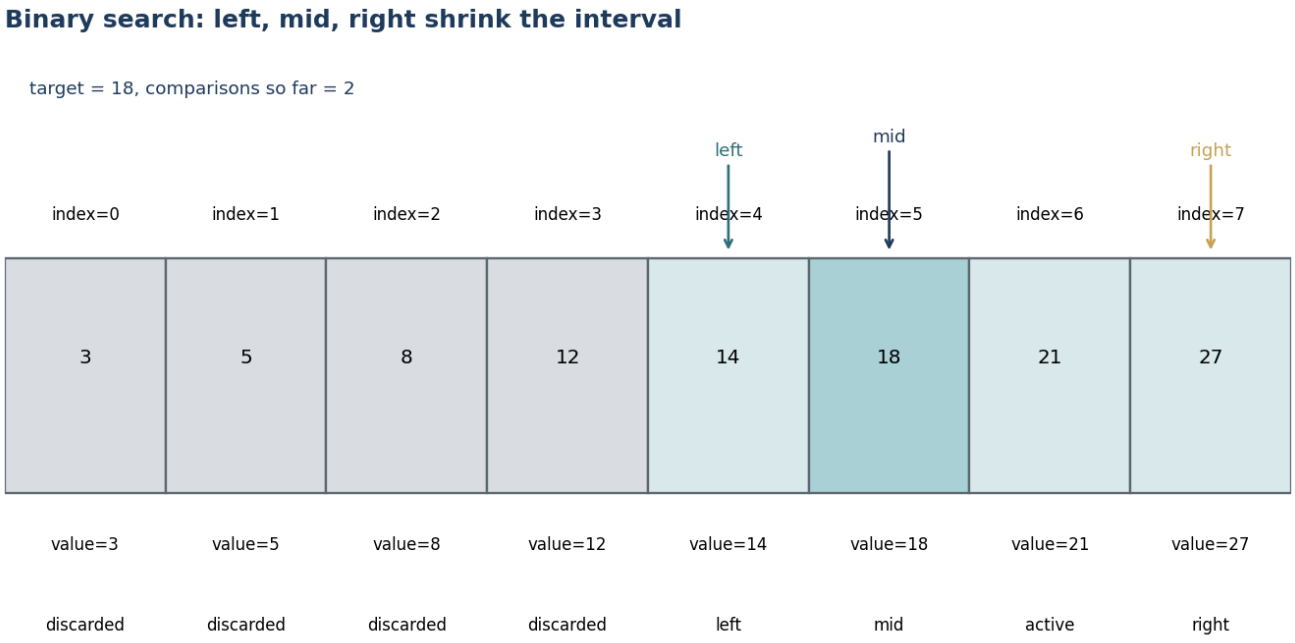
```
def linear_search(data, target, *, count_ops=False):
    """Scan the list from left to right until target is found.

    Parameters
    ----------
    data:
        The list being searched.
    target:
        The value we want to find.
    count_ops:
        If True, also return the number of equality comparisons made.

    Returns
    -------
    index or (index, comparisons)
        index is the location of target, or -1 if target is absent.
        comparisons counts how many values were inspected.
```

```
␣␣␣␣"""
   comparisons = 0
   for index, value in enumerate(data):
      # index marks the current position in the list.
      # value is the item stored at that position.
      comparisons += 1
      if value == target:
         return (index, comparisons) if count_ops else index
   return (-1, comparisons) if count_ops else -1


assert linear_search([3, 5, 8, 12, 14, 18, 21, 27], 18) == 5

demo_search_data = [3, 5, 8, 12, 14, 18, 21, 27]
target = 18

plot_linear_search_demo(
   demo_search_data,
   target,
   title='Linear␣search:␣one␣moving␣index',
)
plt.show()
```




```
def binary_search(data, target, *, count_ops=False):
   """Search␣a␣sorted␣list␣by␣repeatedly␣halving␣the␣active␣interval.

␣␣␣␣left␣and␣right␣mark␣the␣current␣inclusive␣search␣window.
␣␣␣␣mid␣is␣the␣position␣inspected␣in␣the␣middle␣of␣that␣window.
␣␣␣␣"""
   left = 0
   right = len(data) - 1
   comparisons = 0

   while left <= right:
      mid = (left + right) // 2
      comparisons += 1

      if data[mid] == target:
         return (mid, comparisons) if count_ops else mid

      # If the midpoint value is too small, the target can only be on the right.
      if data[mid] < target:
         left = mid + 1
      else:
```

```
        right = mid - 1

    return (-1, comparisons) if count_ops else -1


assert binary_search([3, 5, 8, 12, 14, 18, 21, 27], 18) == 5

demo_search_data = [3, 5, 8, 12, 14, 18, 21, 27]
target = 18

plot_binary_search_demo(
    demo_search_data,
    target,
    title='Binary␣search:␣left,␣mid,␣right␣shrink␣the␣interval',
)
plt.show()
```

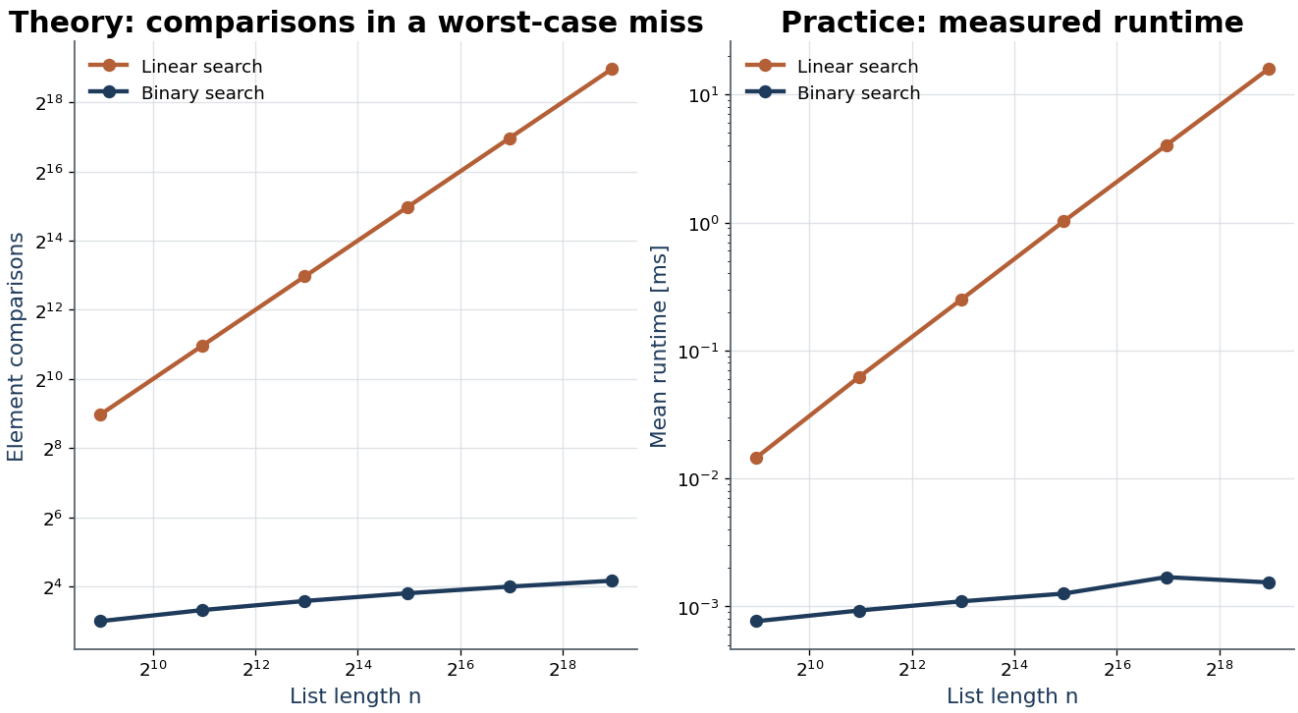


The two search diagrams now stand on their own: linear search highlights one moving `index`, while binary search highlights the shrinking interval described by `left`, `mid`, and `right`.

```
# We count operations on worst-case misses because the pattern is very clean:
# both algorithms must fully commit to their search strategy.
search_sizes = np.array([500, 2_000, 8_000, 32_000, 128_000, 512_000])
sorted_search_data = {int(size): list(range(int(size))) for size in search_sizes}
missing_target = -1

def timed_linear_search(size):
    """Time␣linear␣search␣on␣a␣list␣of␣the␣requested␣size."""
    return linear_search(sorted_search_data[int(size)], missing_target)

def timed_binary_search(size):
    """Time␣binary␣search␣on␣the␣same␣kind␣of␣data."""
    return binary_search(sorted_search_data[int(size)], missing_target)

search_story_rows = []
for size in search_sizes:
    _, linear_ops = linear_search(sorted_search_data[int(size)], missing_target, count_ops=True)
    _, binary_ops = binary_search(sorted_search_data[int(size)], missing_target, count_ops=True)
    search_story_rows.append(
        {
            'n': int(size),
            'linear␣search␣comparisons': linear_ops,
            'binary␣search␣comparisons': binary_ops,
        }
```

```
    )
search_story_df = pd.DataFrame(search_story_rows)

# Binary search is so fast here that we repeat it more times to get a stable mean.
linear_search_timing = time_size_series(timed_linear_search, search_sizes, repeat=7, warmup=1)
binary_search_timing = time_size_series(timed_binary_search, search_sizes, repeat=31, warmup=5)

search_story_df['linear␣mean␣ms'] = linear_search_timing['mean_s'] * 1_000
search_story_df['binary␣mean␣ms'] = binary_search_timing['mean_s'] * 1_000
display_wrapped(search_story_df)

fig, axes = lecture_figure(kind='wide', ncols=2)

axes[0].plot(
    search_story_df['n'],
    search_story_df['linear␣search␣comparisons'],
    marker='o',
    label='Linear␣search',
    color=COLORS['rust'],
)
axes[0].plot(
    search_story_df['n'],
    search_story_df['binary␣search␣comparisons'],
    marker='o',
    label='Binary␣search',
    color=COLORS['navy'],
)
axes[0].set_xscale('log', base=2)
axes[0].set_yscale('log', base=2)
axes[0].set_title('Theory:␣comparisons␣in␣a␣worst-case␣miss')
axes[0].set_xlabel('List␣length␣n')
axes[0].set_ylabel('Element␣comparisons')
axes[0].legend(loc='upper␣left')

axes[1].plot(
    search_story_df['n'],
    search_story_df['linear␣mean␣ms'],
    marker='o',
    label='Linear␣search',
    color=COLORS['rust'],
)
axes[1].plot(
    search_story_df['n'],
    search_story_df['binary␣mean␣ms'],
    marker='o',
    label='Binary␣search',
    color=COLORS['navy'],
)
axes[1].set_xscale('log', base=2)
axes[1].set_yscale('log', base=10)
axes[1].set_title('Practice:␣measured␣runtime')
axes[1].set_xlabel('List␣length␣n')
axes[1].set_ylabel('Mean␣runtime␣[ms]')
axes[1].legend(loc='upper␣left')

plt.show()
```

	n	linear search comparisons	binary search comparisons	linear mean ms	binary mean ms
0	500	500	8	0.0146	0.0008
1	2000	2000	10	0.0620	0.0009
2	8000	8000	12	0.2503	0.0011
3	32000	32000	14	1.0226	0.0013
4	128000	128000	16	4.0131	0.0017
5	512000	512000	18	15.8601	0.0015



The theoretical picture is clean: linear search grows in proportion to `n`, while binary search grows in proportion to `log n`.

The timing picture tells the same story, but with real-world caveats: Python overhead, cache effects, and measurement noise all matter. Timings help us estimate practical cost; operation counts help us explain **why** the trend appears.

```
display_wrapped(Markdown(markdown_box(
    'If␣the␣data␣is␣already␣sorted,␣binary␣search␣changes␣the␣scaling␣story␣completely:␣halving␣the␣remaining␣interval␣
        beats␣scanning␣every␣element.'
)))

display_wrapped(Markdown(instructor_prompt(
    'Ask␣whether␣binary␣search␣is␣always␣better.␣Students␣should␣notice␣the␣hidden␣assumption:␣the␣data␣must␣already␣be␣
        sorted.'
)))
```

> **Takeaway:** If the data is already sorted, binary search changes the scaling story completely: halving the remaining interval beats scanning every element.
>
> **Instructor Prompt:** Ask whether binary search is always better. Students should notice the hidden assumption: the data must already be sorted.

3. Sorting as a scaling lesson

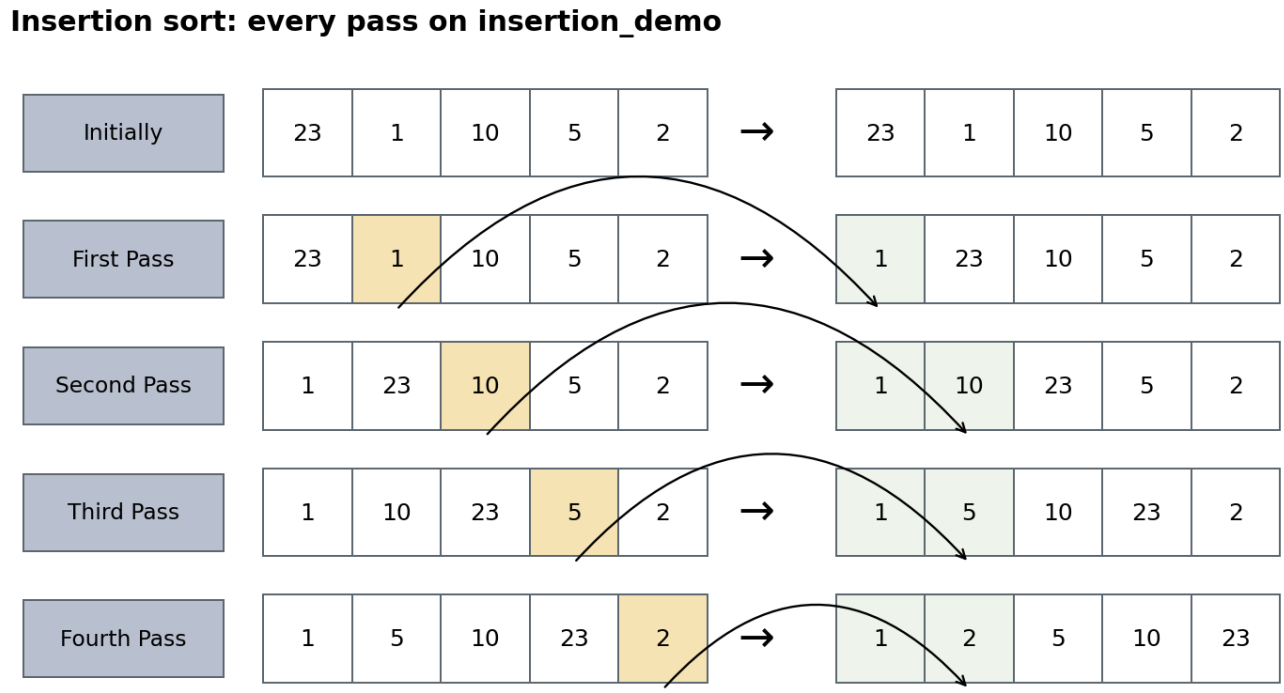
Sorting is richer than search because the algorithm has to reorganize the whole list.

We will compare: - **insertion sort**, a simple `O(n^2)` method that is easy to explain - **merge sort**, a classic `O(n log n)` divide-and-conquer method

Sorting makes the difference sharper because two algorithms solve the same task while organizing work very differently. The implementations here stay educational on purpose, so the visual story remains readable during live lecture.

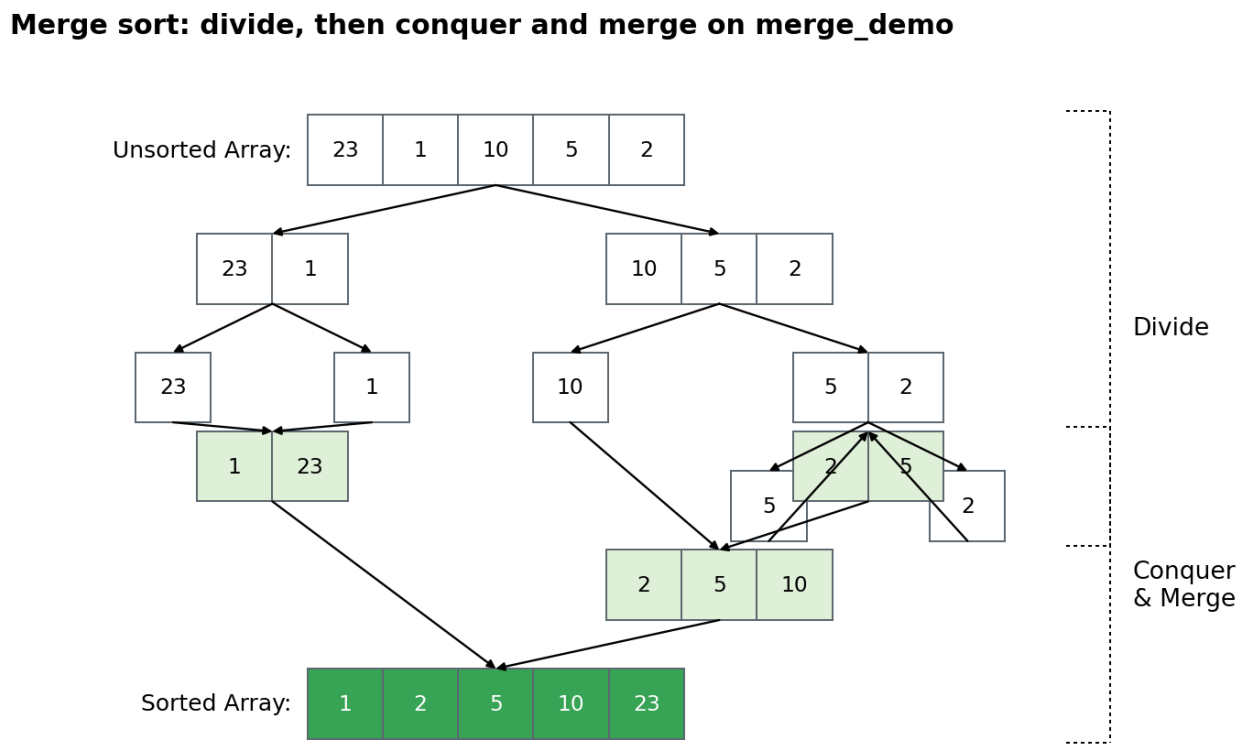
```
def insertion_sort(values):
    """Sort␣by␣growing␣a␣sorted␣prefix␣one␣position␣at␣a␣time.

␣␣␣␣i␣marks␣the␣first␣element␣not␣yet␣inserted␣into␣the␣sorted␣prefix.
␣␣␣␣key␣stores␣the␣element␣currently␣being␣inserted.
␣␣␣␣j␣walks␣left␣through␣the␣sorted␣prefix␣while␣larger␣values␣are␣shifted.
␣␣␣␣"""
    data = list(values)
    comparisons = 0

    for i in range(1, len(data)):
        # Treat data[:i] as the sorted prefix built so far.
        key = data[i]
        j = i - 1
```

```
        while j >= 0:
            # Compare key against the prefix from right to left.
            comparisons += 1
            if data[j] > key:
                # Shift larger values one step to the right to open a slot.
                data[j + 1] = data[j]
                j -= 1
            else:
                # Stop once key has found the correct position.
                break

        # Insert key into the gap left behind by the shifts.
        data[j + 1] = key
    return data, comparisons


sample = rng.integers(0, 50, size=12).tolist()
assert insertion_sort(sample)[0] == sorted(sample)

insertion_demo = [23, 1, 10, 5, 2]

plot_insertion_sort_passes(
    insertion_demo,
    title='Insertion␣sort:␣every␣pass␣on␣insertion_demo',
)
plt.show()
```




```
def merge_sort(values):
    """Sort␣by␣recursively␣splitting␣the␣list,␣then␣merging␣sorted␣halves.

␣␣␣␣left␣and␣right␣are␣the␣sorted␣sublists␣produced␣by␣recursion.
␣␣␣␣i␣and␣j␣track␣the␣next␣unread␣positions␣in␣those␣sublists.
␣␣␣␣merged␣stores␣the␣output␣built␣so␣far.
␣␣␣␣"""
    data = list(values)
    if len(data) <= 1:
        # A list of length 0 or 1 is already sorted.
        return data, 0

    # Split the current problem into two smaller sorting problems.
    middle = len(data) // 2
    left, left_comparisons = merge_sort(data[:middle])
```

```
    right, right_comparisons = merge_sort(data[middle:])

    # Merge the two sorted halves back into one sorted output list.
    merged = []
    i = 0
    j = 0
    comparisons = left_comparisons + right_comparisons

    while i < len(left) and j < len(right):
        # Compare the next unread value from each half.
        comparisons += 1
        if left[i] <= right[j]:
            # Take the smaller value and advance that pointer.
            merged.append(left[i])
            i += 1
        else:
            merged.append(right[j])
            j += 1

    # One half may still contain leftover values; append them in order.
    merged.extend(left[i:])
    merged.extend(right[j:])
    return merged, comparisons


sample = rng.integers(0, 50, size=12).tolist()
assert merge_sort(sample)[0] == sorted(sample)

merge_demo = [23, 1, 10, 5, 2]

plot_merge_sort_decomposition(
    merge_demo,
    title='Merge␣sort:␣divide,␣then␣conquer␣and␣merge␣on␣merge_demo',
)
plt.show()
```



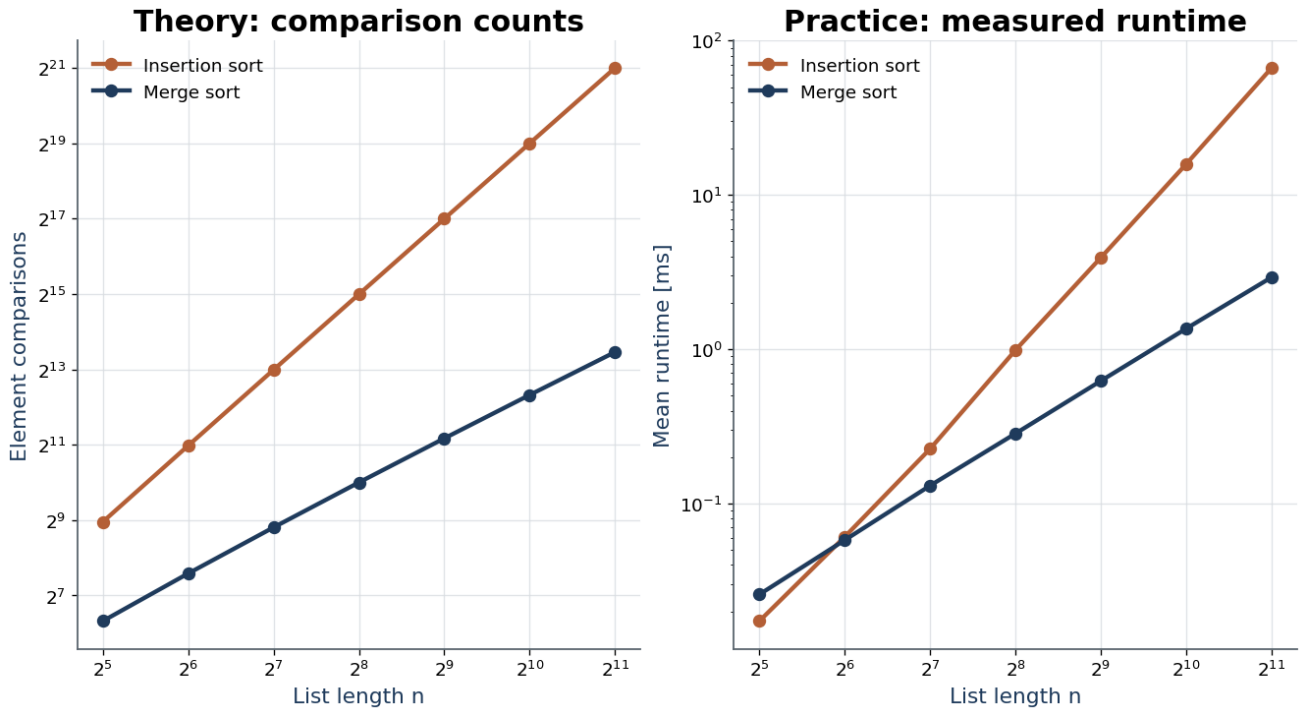
The sorting visuals now separate the two ideas cleanly: insertion sort shows every pass on one array, while merge sort shows the full recursive split-and-merge structure for the same example.

```
# Reversed input gives insertion sort a very clear worst-case comparison count.
sort_sizes = np.array([32, 64, 128, 256, 512, 1_024, 2_048])
```

```
sort_inputs = {
   int(size): rng.integers(0, 10 * int(size), size=int(size)).tolist()
   for size in sort_sizes
}

def timed_insertion_sort(size):
   """Time␣insertion␣sort␣on␣one␣fixed␣random␣list␣for␣this␣size."""
   return insertion_sort(sort_inputs[int(size)])[0]

def timed_merge_sort(size):
   """Time␣merge␣sort␣on␣the␣same␣input␣family."""
   return merge_sort(sort_inputs[int(size)])[0]

sort_story_rows = []
for size in sort_sizes:
   reversed_data = list(range(int(size), 0, -1))
   _, insertion_comparisons = insertion_sort(reversed_data)
   _, merge_comparisons = merge_sort(reversed_data)
   sort_story_rows.append(
      {
         'n': int(size),
         'insertion␣sort␣comparisons': insertion_comparisons,
         'merge␣sort␣comparisons': merge_comparisons,
      }
   )
sort_story_df = pd.DataFrame(sort_story_rows)

insertion_timing = time_size_series(timed_insertion_sort, sort_sizes, repeat=3, warmup=1)
merge_timing = time_size_series(timed_merge_sort, sort_sizes, repeat=5, warmup=1)

sort_story_df['insertion␣mean␣ms'] = insertion_timing['mean_s'] * 1_000
sort_story_df['merge␣mean␣ms'] = merge_timing['mean_s'] * 1_000
display_wrapped(sort_story_df)

fig, axes = lecture_figure(kind='wide', ncols=2)

axes[0].plot(
   sort_story_df['n'],
   sort_story_df['insertion␣sort␣comparisons'],
   marker='o',
   label='Insertion␣sort',
   color=COLORS['rust'],
)
axes[0].plot(
   sort_story_df['n'],
   sort_story_df['merge␣sort␣comparisons'],
   marker='o',
   label='Merge␣sort',
   color=COLORS['navy'],
)
axes[0].set_xscale('log', base=2)
axes[0].set_yscale('log', base=2)
axes[0].set_title('Theory:␣comparison␣counts')
axes[0].set_xlabel('List␣length␣n')
axes[0].set_ylabel('Element␣comparisons')
axes[0].legend(loc='upper␣left')

axes[1].plot(
   sort_story_df['n'],
   sort_story_df['insertion␣mean␣ms'],
   marker='o',
   label='Insertion␣sort',
   color=COLORS['rust'],
)
axes[1].plot(
   sort_story_df['n'],
   sort_story_df['merge␣mean␣ms'],
   marker='o',
   label='Merge␣sort',
```

```
    color=COLORS['navy'],
)
axes[1].set_xscale('log', base=2)
axes[1].set_yscale('log', base=10)
axes[1].set_title('Practice:␣measured␣runtime')
axes[1].set_xlabel('List␣length␣n')
axes[1].set_ylabel('Mean␣runtime␣[ms]')
axes[1].legend(loc='upper␣left')

plt.show()
```

	n	insertion sort comparisons	merge sort comparisons	insertion mean ms	merge mean ms
0	32	496	80	0.0173	0.0258
1	64	2016	192	0.0609	0.0579
2	128	8128	448	0.2262	0.1307
3	256	32640	1024	0.9869	0.2843
4	512	130816	2304	3.9247	0.6236
5	1024	523776	5120	15.8612	1.3608
6	2048	2096128	11264	66.5667	2.9265



Both sorts solve the same problem, but they do not scale the same way. The simple algorithm is attractive on the board because it is easy to read. The more efficient algorithm wins as soon as the list becomes moderately large.

```
display_wrapped(Markdown(markdown_box(
    'Sorting␣is␣where␣asymptotic␣language␣becomes␣concrete:␣an␣O(n^2)␣method␣can␣look␣harmless␣on␣32␣items␣and␣painful␣
        on␣2,048.'
)))

display_wrapped(Markdown(instructor_prompt(
    'Ask␣students␣whether␣the␣faster␣algorithm␣would␣still␣be␣worth␣it␣for␣5␣elements.␣This␣opens␣the␣door␣to␣constants
        ,␣simplicity,␣and␣real␣engineering␣tradeoffs.'
)))
```

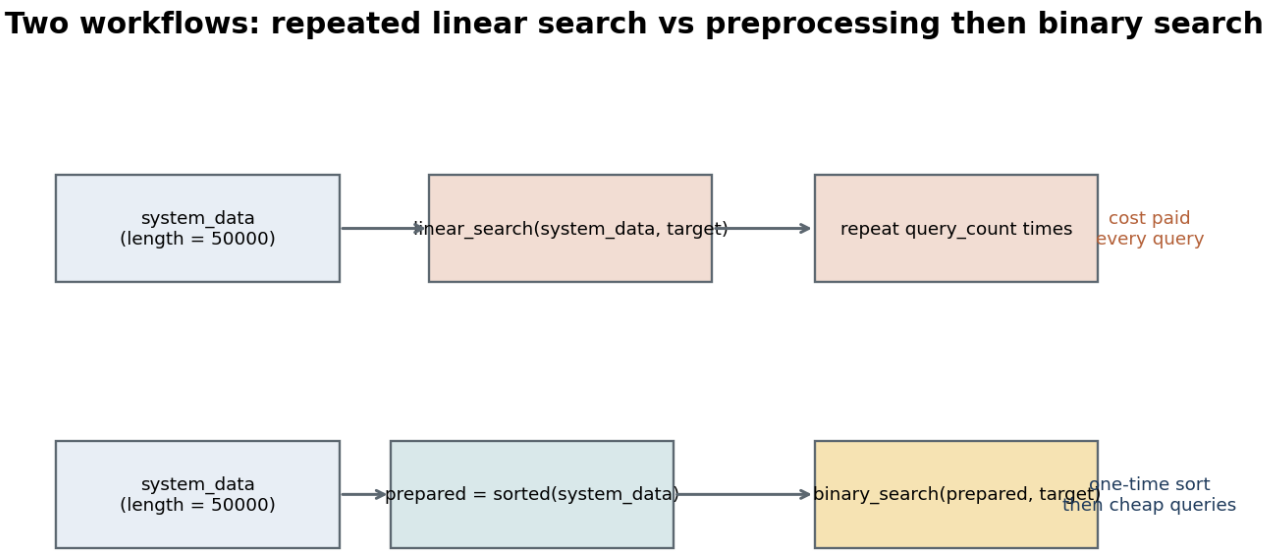
> **Takeaway:** Sorting is where asymptotic language becomes concrete: an O(n^2) method can look harmless on 32 items and painful on 2,048.
>
> **Instructor Prompt:** Ask students whether the faster algorithm would still be worth it for 5 elements. This opens the door to constants, simplicity, and real engineering tradeoffs.

4. Why preprocessing matters

Binary search is fast **after** the data is sorted. That means the full system-level question is not just "Which search is faster?" but rather "What is the total cost of preparing the data and then answering many queries?"

Preprocessing changes the system-level question. An expensive setup step may still be the right decision if it unlocks much cheaper repeated queries later on.

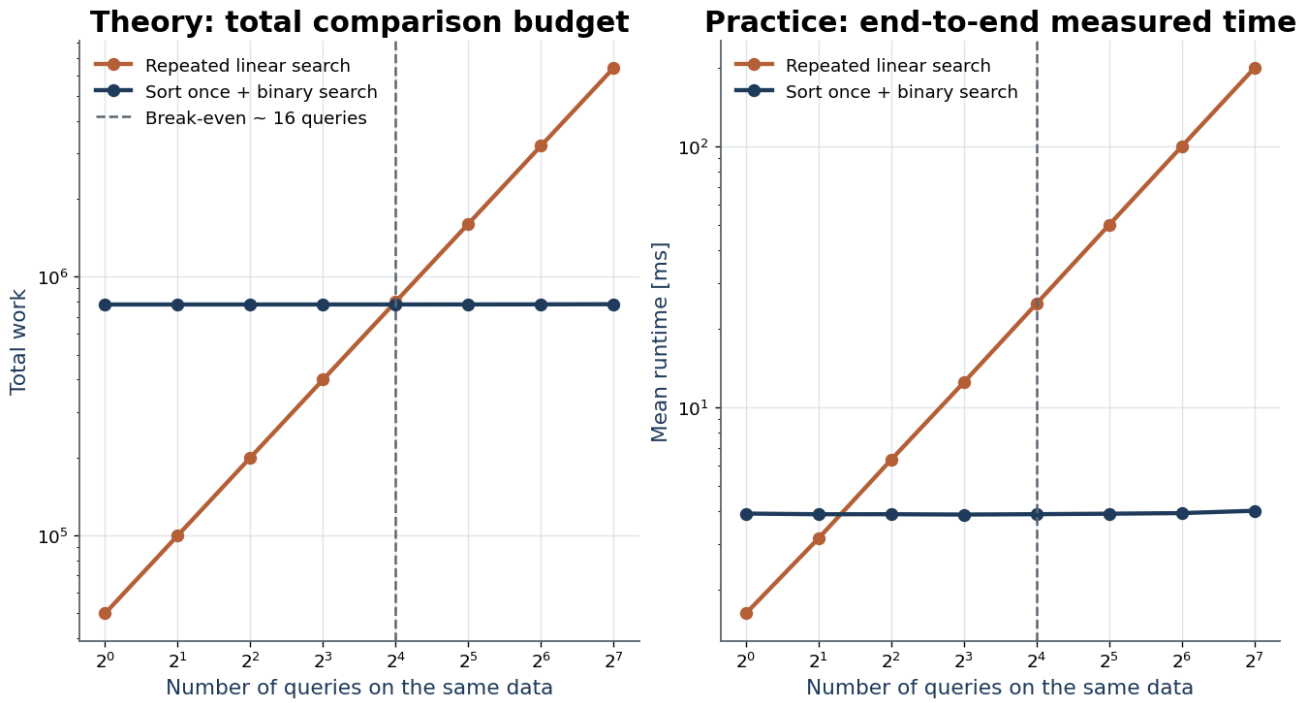
```
system_n = 50_000
query_counts = np.array([1, 2, 4, 8, 16, 32, 64, 128])

plot_preprocessing_workflow(system_n)
plt.show()
```




```
system_data = rng.permutation(system_n).tolist()
system_target = -1

# A coarse theory model for repeated queries.
estimated_sort_cost = system_n * math.log2(system_n)
estimated_binary_cost = math.ceil(math.log2(system_n))
break_even_queries = math.ceil(
   estimated_sort_cost / (system_n - estimated_binary_cost)
)

preprocessing_df = pd.DataFrame(
   {
      'queries': query_counts,
      'linear␣total␣comparisons': query_counts * system_n,
      'sort␣once␣+␣binary␣comparisons': estimated_sort_cost
      + query_counts * estimated_binary_cost,
   }
)

def repeated_linear_queries(query_count):
   """Run␣query_count␣independent␣linear␣misses␣on␣the␣same␣unsorted␣data."""
   for _ in range(int(query_count)):
      linear_search(system_data, system_target)
   return -1

def sort_once_then_binary_queries(query_count):
   """Pay␣one␣sorting␣cost,␣then␣reuse␣the␣prepared␣list␣for␣all␣queries."""
   prepared = sorted(system_data)
   for _ in range(int(query_count)):
      binary_search(prepared, system_target)
   return -1

linear_system_timing = time_size_series(repeated_linear_queries, query_counts, repeat=5, warmup=1)
sort_then_search_timing = time_size_series(sort_once_then_binary_queries, query_counts, repeat=5, warmup=1)
```

```
preprocessing_story_df = preprocessing_df.copy()
preprocessing_story_df['linear␣mean␣ms'] = linear_system_timing['mean_s'] * 1_000
preprocessing_story_df['sort␣once␣+␣binary␣mean␣ms'] = sort_then_search_timing['mean_s'] * 1_000
display_wrapped(preprocessing_story_df)

fig, axes = lecture_figure(kind='wide', ncols=2)

axes[0].plot(
   preprocessing_story_df['queries'],
   preprocessing_story_df['linear␣total␣comparisons'],
   marker='o',
   label='Repeated␣linear␣search',
   color=COLORS['rust'],
)
axes[0].plot(
   preprocessing_story_df['queries'],
   preprocessing_story_df['sort␣once␣+␣binary␣comparisons'],
   marker='o',
   label='Sort␣once␣+␣binary␣search',
   color=COLORS['navy'],
)
axes[0].axvline(
   break_even_queries,
   linestyle='--',
   linewidth=1.5,
   color=COLORS['gray'],
   label=f'Break-even␣~␣{break_even_queries}␣queries',
)
axes[0].set_xscale('log', base=2)
axes[0].set_yscale('log', base=10)
axes[0].set_title('Theory:␣total␣comparison␣budget')
axes[0].set_xlabel('Number␣of␣queries␣on␣the␣same␣data')
axes[0].set_ylabel('Total␣work')
axes[0].legend(loc='upper␣left')

axes[1].plot(
   preprocessing_story_df['queries'],
   preprocessing_story_df['linear␣mean␣ms'],
   marker='o',
   label='Repeated␣linear␣search',
   color=COLORS['rust'],
)
axes[1].plot(
   preprocessing_story_df['queries'],
   preprocessing_story_df['sort␣once␣+␣binary␣mean␣ms'],
   marker='o',
   label='Sort␣once␣+␣binary␣search',
   color=COLORS['navy'],
)
axes[1].axvline(
   break_even_queries,
   linestyle='--',
   linewidth=1.5,
   color=COLORS['gray'],
)
axes[1].set_xscale('log', base=2)
axes[1].set_yscale('log', base=10)
axes[1].set_title('Practice:␣end-to-end␣measured␣time')
axes[1].set_xlabel('Number␣of␣queries␣on␣the␣same␣data')
axes[1].set_ylabel('Mean␣runtime␣[ms]')
axes[1].legend(loc='upper␣left')

plt.show()
```

	queries	linear total comparisons	sort once + binary comparisons	linear mean ms	sort once + binary mean ms
0	1	50000	780,498.0237	1.6247	3.9202
1	2	100000	780,514.0237	3.1529	3.8966
2	4	200000	780,546.0237	6.3093	3.8979
3	8	400000	780,610.0237	12.4804	3.8863
4	16	800000	780,738.0237	24.9848	3.8989
5	32	1600000	780,994.0237	50.0731	3.9155
6	64	3200000	781,506.0237	100.0547	3.9363
7	128	6400000	782,530.0237	199.8366	4.0192



For a single lookup, sorting first is usually wasteful. For many lookups on the same dataset, preprocessing can be the right strategic move.

This is a useful complexity habit: always ask whether you are paying a one-time setup cost or a cost that repeats for every query.

```
display_wrapped(Markdown(markdown_box(
    'Asymptotic␣complexity␣is␣often␣a␣systems␣story,␣not␣just␣an␣algorithm␣story.␣Data␣preparation␣can␣make␣a␣slower-
        looking␣method␣win␣over␣repeated␣use.'
)))

display_wrapped(Markdown(instructor_prompt(
    'Ask␣students␣where␣they␣see␣this␣tradeoff␣in␣practice:␣indexing␣a␣database,␣building␣a␣lookup␣table,␣or␣
        precomputing␣simulation␣data.'
)))
```

> **Takeaway:** Asymptotic complexity is often a systems story, not just an algorithm story. Data preparation can make a slower-looking method win over repeated use.
>
> **Instructor Prompt:** Ask students where they see this tradeoff in practice: indexing a database, building a lookup table, or precomputing simulation data.

5. Recap / takeaways

```
display_wrapped(Markdown(markdown_box(
    """
␣␣␣␣1.␣Input␣size␣n␣has␣to␣be␣defined␣carefully.␣In␣this␣notebook␣it␣meant␣list␣length.
␣␣␣␣2.␣Growth␣laws␣that␣look␣similar␣on␣small␣examples␣separate␣dramatically␣as␣n␣increases.
␣␣␣␣3.␣Operation␣counts␣explain␣the␣scaling␣law;␣timing␣experiments␣reveal␣practical␣constants␣and␣overhead.
␣␣␣␣4.␣Binary␣search␣is␣powerful␣only␣after␣preprocessing␣makes␣the␣data␣searchable␣in␣that␣way.
␣␣␣␣5.␣Sorting␣and␣searching␣are␣not␣isolated␣tasks:␣the␣total␣workflow␣matters.
␣␣␣␣""",
    title='Recap',
```

```
)))
```

Recap: 1. Input size n has to be defined carefully. In this notebook it meant list length. 2. Growth laws that look similar on small examples separate dramatically as n increases. 3. Operation counts explain the scaling law; timing experiments reveal practical constants and overhead. 4. Binary search is powerful only after preprocessing makes the data searchable in that way. 5. Sorting and searching are not isolated tasks: the total workflow matters.

6. Questions for students / lecture prompts

- If `n` doubles, which of today's growth laws changes the least? Which changes the most?
- When does a simple algorithm still make sense, even if its asymptotic class is worse?
- Why is binary search not a fair comparison against linear search unless we talk about sorted input?
- Can you name a real task where paying a preprocessing cost once is clearly worth it?
- If timing and theory disagree slightly on a laptop, which result should guide your conceptual understanding?